Trigonometry & Polar Coordinates

ANSWER KEY

Section 1: Product-to-Sum & Sum-to-Product Identities

1 Express $2\sin(5x)\cos(3x)$ as a sum of trigonometric functions.

a. $\sin(8x) + \sin(2x)$ ✓

b. $\sin(8x) - \sin(2x)$

c. $\cos(8x) + \cos(2x)$

d. $\cos(8x) - \cos(2x)$

Using the product-to-sum identity $2\sin A\cos B = \sin(A+B) + \sin(A-B)$ gives $\sin(8x) + \sin(2x)$.

2 Simplify the expression $\frac{\sin(3x) + \sin(x)}{\cos(3x) + \cos(x)}$.

a. $\tan(2x)$ ✓

b. $\tan(x)$

c. $\cot(2x)$

d. $\tan(4x)$

Using sum-to-product: numerator $= 2\sin(2x)\cos(x)$ and denominator $= 2\cos(2x)\cos(x)$. Dividing gives $\frac{\sin(2x)}{\cos(2x)} = \tan(2x)$.

3 What is the exact value of $\cos(105^\circ) - \cos(15^\circ)$?

a. $-\frac{\sqrt{2}}{2}$ ✓

b. $\frac{\sqrt{2}}{2}$

c. $-\frac{\sqrt{6}}{2}$

d. $\frac{\sqrt{6}}{2}$

Sum-to-product: $-2\sin(60^\circ)\sin(45^\circ) = -2(\frac{\sqrt{3}}{2})(\frac{\sqrt{2}}{2}) = -\frac{\sqrt{6}}{2}$.

Section 2: Polar Curves & Graphs

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4 What geometric curve is represented by the polar equation $r = 2 + 2\cos\theta$?

a. **Cardioid** ✓

b. Limaçon with inner loop

c. Three-petal rose

d. Circle

An equation of the form $r = a + a\cos\theta$ has equal coefficients ($a = b$), which represents a cardioid (heart-shaped curve).

5 How many petals does the rose curve $r = 4\cos(3\theta)$ have?

a. 3 ✓

b. 6

c. 4

d. 8

For a rose curve $r = a\cos(n\theta)$, if n is odd, the curve has n petals. Here $n = 3$, so it has 3 petals.

6 How many petals does the rose curve $r = 2\sin(4\theta)$ have?

a. 4

b. 8 ✓

c. 16

d. 2

If n is even in $r = a\sin(n\theta)$, the curve has $2n$ petals. Here $n = 4$, so it has $2(4) = 8$ petals.

Section 3: Polar to Rectangular Conversions

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7 Convert the polar point $(4, \frac{5\pi}{6})$ to rectangular coordinates (x, y).

a. $(-2\sqrt{3}, 2)$ ✓

b. $(2\sqrt{3}, -2)$

c. $(-2, 2\sqrt{3})$

d. $(2, -2\sqrt{3})$

$x = r\cos\theta = 4\cos(\frac{5\pi}{6}) = 4(-\frac{\sqrt{3}}{2}) = -2\sqrt{3}$. $y = r\sin\theta = 4\sin(\frac{5\pi}{6}) = 4(\frac{1}{2}) = 2$.

8 Convert the rectangular point $(-3, -3)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(3\sqrt{2}, \frac{5\pi}{4})$ ✓

b. $(3\sqrt{2}, \frac{3\pi}{4})$

c. $(6, \frac{5\pi}{4})$

d. $(3\sqrt{2}, \frac{7\pi}{4})$

$r = \sqrt{(-3)^2 + (-3)^2} = \sqrt{18} = 3\sqrt{2}$. Point is in Quadrant III: $\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$.

9 What is the Cartesian equation for the polar line $\theta = \frac{\pi}{3}$?

a. $y = \sqrt{3}x$ ✓

b. $y = \frac{\sqrt{3}}{3}x$

c. $y = x + \frac{\pi}{3}$

d. $x = \sqrt{3}y$

Take tangent of both sides: $\tan\theta = \tan(\frac{\pi}{3}) = \sqrt{3}$. Since $\tan\theta = \frac{y}{x}$, we have $\frac{y}{x} = \sqrt{3} \Rightarrow y = \sqrt{3}x$.

Section 4: Trigonometric Equations on Restricted Domains

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10 Solve $\cos(3x) = \frac{1}{2}$ for $x \in [0, \pi]$.

a. $\frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$ ✓

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{9}, \frac{7\pi}{9}$

d. $\frac{\pi}{3}, \frac{2\pi}{3}$

For $x \in [0, \pi]$, $3x \in [0, 3\pi]$. $\cos(3x) = 1/2 \Rightarrow 3x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$. Dividing by 3 gives $x = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$.

11 Find all solutions to $\sin(2x) = \cos(x)$ in the interval $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ ✓

b. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$

c. $\frac{\pi}{3}, \frac{2\pi}{3}$

d. $\frac{\pi}{6}, \frac{5\pi}{6}$

$2\sin x\cos x - \cos x = 0 \Rightarrow \cos x(2\sin x - 1) = 0$. $\cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}$. $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$.

12 What is the range of $f(x) = 2 - 3\sin(4x)$?

a. $[-1, 5]$ ✓

b. $[-3, 3]$

c. $[-5, 1]$

d. $[2, 5]$

Since $-1 \le \sin(4x) \le 1$, we have $-3 \le -3\sin(4x) \le 3$. Adding 2 gives $-1 \le f(x) \le 5$.

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